ADVANCED BIOFABRICATION STRATEGIES FOR TISSUE REGENERATION

Melt Electrowriting of Polymeric Scaffolds for Rotator Cuff Enthesis Repair

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1. Lay Summary and Abstract

1.1 Lay Summary

Tendon injuries in the shoulder, especially to the rotator cuff, are common and can seriously affect movement and quality of life. Unfortunately, current surgical treatments often fail to heal properly, especially at the point where the tendon attaches to bone. This project explored how to build better support structures – known as scaffolds – that can help guide healing in this area.

We used a sustainable and biocompatible material called polyhydroxyalkanoate (PHA), which is known to be safe for the body, and tried to print it in very thin, controlled layers using a method called melt electrowriting, a bioprinting technique similar to 3D printing, except a high voltage is applied to stretch out the melt from the nozzle, achieving fibre diameters in the micrometre range (10^{-6} m). To make sure the printed fibres were the right size and shape, we developed custom software to analyse them from microscope images.

Through extensive testing, we found a specific set of conditions that produced the most accurate and stable structures. These findings could help future researchers develop more efficient scaffolds for tendon repair and improve long-term recovery outcomes after injury.

1.2 Abstract

Rotator cuff tendon injuries are common and challenging to repair due to the complexity of the enthesis — the transitional region connecting tendon to bone. Traditional surgical approaches often fail to regenerate this graded interface effectively, leading to high retear rates. With a high prevalence in the population, occurring between 6.8% and 22.4% of individuals above the age of 40 (Kuhn, 2023), and large hindrance on quality of life, it warrants further investigation and study into effective treatment strategies. This project investigates the development of engineered scaffolds, a promising solution that can support tendon healing by mimicking the structure of the enthesis.

Polyhydroxyalkanoates (PHAs), specifically medium-chain length PHA (MCL-PHA), was chosen as the scaffold material due to their biocompatibility, biodegradability, and

sustainable origins, offering key advantages over the widely used polycaprolactone (PCL). The goal was to optimise a precise and controlled bioprinting technique, melt electrowriting (MEW), to fabricate PHA fibre scaffolds.

The project followed three phases: establishing benchmark MEW performance and print quality using PCL, adapting and optimising settings for MCL-PHA, and testing the reliability of optimised parameters with gradient scaffold patterns. A custom Python-based image analysis program was developed to measure fibre diameters from microscope images, allowing quantitative assessment of fibre diameters and enabling objective evaluation of print quality through Design of Experiments (DOE).

Two major outcomes were observed: lower-temperature prints which were intensively studied and optimised, showed thin fibres with high print consistency but suffered from fibre merging, resulting in larger diameters. Higher-temperature settings produced significantly improved scaffold morphology with thin, layered fibres, and improved printing stability, producing the highest-quality PHA prints achieved in the study. These findings provide a foundation for further biological testing and support future work in high-resolution printing, scaffold fidelity analysis, and advanced pattern generation for biomimetic tissue design.

Keywords: Rotator Cuff Disease (RCD), Enthesis, Melt Electrowriting (MEW), Polyhydroxyalkanoate (PHA), Polycaprolactone (PCL), Design of Experiments (DOE), Scanning Electron Microscope (SEM)

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2. Introduction

2.1 Current Challenges in Tendon Repair

Rotator cuff injuries are one of the most prevalent musculoskeletal disorders, particularly affecting the aging population. Studies suggest that rotator cuff tears can occur between 6.8% and 22.4% of individuals above the age of 40, with the prevalence increasing with age and repetitive overhead activity (Kuhn, 2023). These injuries often lead to persistent pain, weakness, and loss of shoulder function, significantly impairing quality of life (Varacallo and Mair, 2019). With such a high prevalence in the population and severe impact on quality of life, it warrants further investigation and study into effective treatment strategies.

Conventional treatment strategies, including surgical repair using sutures, autografts, or synthetic materials, frequently suffer from high failure rates, particularly due to main site of injury and cause of retear; the tendon-to-bone interface or the enthesis (Lu and Thomopoulos, 2013). High retear rates have been reported due to the poor healing capacity of the enthesis, related to its unique and complex interface and lack of tissue regeneration, causing mechanically inferior scar tissue formation and being unable to regain its original structural and biomechanical properties (Abdalla and Pendegrass, 2023). As such, the difficulty lies not only in reattaching the tendon, but also in regenerating the complex, graded tissue structure of the native enthesis, which transitions from soft tendon to hard bone through a fibrocartilaginous zone. This biological constraint, coupled with the mechanical challenges at the repair site, hinders complete and functional regeneration.

These shortcomings highlight the pressing need for biomimetic scaffolds capable of supporting and guiding cellular regeneration while withstanding mechanical load. The goal is to fabricate scaffolds that resemble the hierarchical organization and mechanical gradients of the rotator cuff enthesis, to promote gradient tissue growth mimicking the natural structure, and improve the durability of surgical repairs.

2.2 Research Statement

The focus of this research is the fabrication and optimization of biodegradable scaffolds using melt electrowriting (MEW) with medium chain length polyhydroxyalkanoates (mcl-PHAs), targeting tendon regeneration in rotator cuff repair. Specifically, the project aims to design scaffolds for the tendon-to-bone interface – the enthesis – using precise, aligned fibre structures that can support directional cell growth and ECM deposition.

While synthetic polymers such as polycaprolactone (PCL) are widely used in tissue engineering due to their mechanical reliability and printability (Arakawa and DeForest, 2017), they suffer from drawbacks including long degradation times and potential inflammatory responses (G. Patricia Sailema-Palate et al., 2016). In contrast, PHAs offer superior biological performance, faster degradation rates aligned with tissue healing timelines and reduced inflammatory risks (Kaniuk and Stachewicz, 2021). However, limited research exists on using PHA with MEW, especially for enthesis-mimetic scaffold applications.

MEW offers a high-resolution platform for scaffold fabrication, enabling tight control over fibre alignment, diameter, and spacing – all critical factors for tendon tissue engineering. In enthesis repair, the challenge lies in replicating the highly complex and unique nature of enthesis, transitioning from anisotropic organised tendon tissue through a fibrocartilaginous zone to bone. Scaffold design must therefore balance biological compatibility with mechanical integrity and offer sufficient control over microarchitecture to influence healing and growth from the cellular level. This requirement for precision and control is why MEW is the preferred choice.

However, MEW is sensitive to several controlled processing parameters: voltage, temperature, pressure, collector distance, and scanning speed, and optimisation for materials like PHA remained a challenge.

This project investigates the feasibility and optimisation of PHA processing via MEW. The research is divided into three phases:

1. Benchmarking MEW performance using PCL, including trial-and-error and Design of Experiments (DOE) approaches.

2. Transitioning to PHA, systematically identifying and optimising parameters that produce thin, layered, and stable prints.
3. Developing and testing custom scaffold patterns, validating the robustness of parameter settings under complex geometries.

As research into PHAs expands, their potential for use in advanced scaffold design becomes more apparent. Continued development of custom polymer blends, functional surface coatings, and tailored chain-length variants could further enhance their suitability for applications (Mehrpouya et al., 2021). Future research can also focus on performing rheology on the scaffold, to determine their mechanical properties, and the long-term in vivo performance of PHA-based scaffolds.

Ultimately, the goal of this project is to develop a MEW protocol for PHA that achieves consistent fibre diameters, stacking capability, and structural fidelity – all necessary conditions for future biological evaluation and clinical scaffold design for rotator cuff repair.

The remainder of the paper is organised into five parts. In section 3, intensive study through literature reviews up until the scope of the project is conducted, covering the anatomy of the enthesis, current treatment methods, scaffolds, and fabrication technologies, and how MEW is used. We will also discuss different materials suitable for application, and advantages to selecting PHA. Section 4 determines the objectives, and Section 5 presents the methodology and materials used. Section 6 exhibits the results with separate discussions for different breakthrough points and challenges. Finally, Section 7 concludes the paper, followed by future work and development.

3. Literature Review & Background Knowledge

3.1 Additive Manufacturing

Additive manufacturing (AM), commonly known as 3D printing, refers to a collection of technologies that build physical objects by layering material according to digital models. Unlike subtractive manufacturing, which removes material from a solid block, AM is a bottom-up approach, reducing waste and enabling the fabrication of complex geometries that are otherwise difficult or impossible to achieve (Kumar, Kumar and Chohan, 2021).

At its core, AM translates a CAD model into sliced layers, each of which is interpreted into toolpaths – most often G-code – guiding the printhead's movement. Depending on the process, materials used can range from thermoplastics and metals to ceramics and even biomaterials (Liu et al., 2019).

In engineering, AM has become a disruptive force across aerospace, automotive, and consumer goods (Rayna and Striukova, 2016). In healthcare, its flexibility has opened the door to patient-specific devices, surgical planning models, and tissue engineering scaffolds (Cui et al., 2012). Its ability to tightly control architecture and material distribution makes it uniquely suited for medical applications where structure and function are tightly linked.

3.2 Biofabrication: Engineering with Biology

Biofabrication is the intersection of biology, materials science, and engineering, defined using additive manufacturing techniques to construct biologically functional products (Harley et al., 2021). Its core objective is to create structures that can support, repair, or mimic living tissue – often by organizing cells and biomaterials into architectures that promote tissue development and integration. Unlike conventional 3D printing, biofabrication requires materials that are biocompatible, and in many cases, bioactive – able to interact positively with living cells (Fonseca et al., 2020).

A major branch in biofabrication is scaffold-based tissue engineering. Scaffolds act as temporary frameworks that provide the mechanical support and structural cues needed for tissue development. They guide cell attachment, proliferation, migration, and

differentiation, mimicking the role of the extracellular matrix (ECM) in native tissues. This is especially critical in musculoskeletal tissues like tendons, where fibre alignment and architecture directly influence mechanical function and cellular behaviour (Liu et al., 2025).

However, biofabrication extends well beyond scaffolds. Techniques such as bioprinting allow for the direct placement of living cells within hydrogels (bioinks), enabling the creation of cell-laden tissues with pre-defined spatial organization. This has enabled advances in producing vascular networks, skin grafts, cartilage structures, and even miniaturized organs-on-chips for drug testing and disease modelling (Fonseca et al., 2020).

These technologies all stem from the same foundation: precise, layer-by-layer control over biologically active materials. What distinguishes biofabrication is not only the resolution of control but also the integration of living components, enabling the creation of functional biological systems rather than just inert models (Kumar, Kumar and Chohan, 2021).

In this project, the focus remains on scaffolds – specifically through melt electrowriting – but the field at large demonstrates how powerful and diverse biofabrication has become in addressing complex biomedical challenges.

3.3 Anatomy and Physiology of the Enthesis

The rotator cuff is a critical musculoskeletal structure responsible for maintaining shoulder stability and enabling a wide range of motion. It is composed of four muscles: the supraspinatus, infraspinatus, subscapularis, and teres minor (Varacallo and Mair, 2019). These muscles converge to form tendinous insertions onto the humeral head, creating a dynamic system that stabilizes the glenohumeral joint during complex arm movements (Burkhart, Esch and R. Scott Jolson, 1993; Clark and Harryman, 1992).

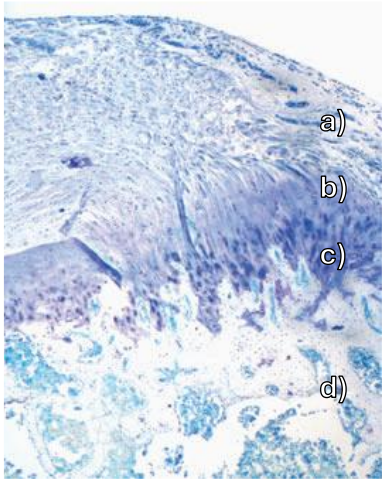

Figure 1: The four zones of the enthesis region.
a) Tendon; b) Unmineralized fibrocartilage; c) Mineralized fibrocartilage; d) Bone (Lu and Thomopoulos, 2013)

Central to rotator cuff function is the enthesis (Figure 1) – the highly specialized tendon-to-bone interface. The enthesis enables the effective distribution of tensile loads from

the flexible, collagen-rich tendon to the rigid mineralized bone. This transition, compositionally categorized into four zones, is structurally continuous to reduce stress concentrations and failure risks, a design critical for sustaining repeated loading cycles (Lu and Thomopoulos, 2013):

a) **Tendon region:** Composed predominantly of type I collagen fibres and fibroblasts arranged in highly aligned bundles, optimized for tensile strength.

b) **Unmineralized fibrocartilaginous zone:** Unmineralized fibrocartilage populated by fibrochondrocytes, mainly composed of type II and III collagen.

c) **Mineralized fibrocartilage:** This region features mainly type II collagen fibres, responsible for bridging soft tissue with hard bone. The border between these two fibrocartilaginous zones is called the "tidemark," representing the calcification border and the mechanical boundary between soft and hard tissues.

d) **Bone:** Bone tissue containing osteoblasts within a mineralized type I collagen matrix, providing the rigid anchor for the entire structure (Thomopoulos et al., 2006; Benjamin et al., 2002).

Each zone is distinguished not only by its biochemical composition but also by mechanical properties. For example, tensile stiffness and modulus increase progressively from tendon to bone.

In the context of rotator cuff disease, it is often either the musculotendinous (muscle-to-tendon) or osteotendinous (tendon-to-bone) junction that fails as degenerative tears where the fibrocartilaginous transition is disrupted, and disorganised scar fibrous tissue with no graded transitional zones is formed at the interface, leaving a higher risk of retear and degeneration (Bonnevie and Mauck, 2018). Repair strategies, therefore, must not simply reconnect the enthesis, but recreate this functional gradient to ensure long-term integration and mechanical resilience (Lu & Thomopoulos, 2013).

From a tissue engineering perspective, this complexity imposes stringent design requirements for scaffolds: not only must they support cell adhesion and growth, but they must also reproduce the microstructural organization, composition, and gradients intrinsic to the native enthesis. Fibre alignment, variable porosity, and spatially

modulated mechanical properties are therefore critical targets in scaffold design aimed at tendon-to-bone repair.

3.4 Current Treatment Options and Scaffolds

Rotator cuff tears vary in severity, and not all require surgical intervention. In many cases, physiotherapy, corticosteroid injections, or rest may be sufficient (Kuhn, 2023). However, for full-thickness tears or those that do not respond to conservative treatment, surgical repair is often necessary. These more complex cases continue to challenge clinicians due to high retear rates and poor regeneration at the tendon-to-bone interface.

Standard surgical techniques involve reattaching the torn tendon to the bone using high-strength sutures. This can be done using suture anchors, which are embedded into the bone, or via transosseous tunnels, where sutures pass through drilled holes in the bone (Derwin et al., 2018). Despite mechanical success in the short term, long-term healing often remains inadequate due to the limited regenerative capacity of the enthesis.

To improve outcomes, surgeons have introduced biological augmentation strategies. These include placing acellular dermal matrix patches over the repair site to reduce stress on the tendon and encourage tissue integration (Bonnevie and Mauck, 2018) or using platelet-rich fibrin (L-PRF) membranes to promote early vascularisation and healing (Zumstein et al., 2014). Techniques such as microfracture of the bone surface (the "crimson duvet" method) aim to stimulate repair by releasing marrow-derived cells and growth factors, though their long-term benefit remains debated (Snyder and Burns, 2009).

On the cellular and molecular front, biologics such as platelet-rich plasma (PRP) deliver concentrated growth factors like PDGF, VEGF, and TGF-β to the injury site. Meta-analyses suggest that PRP can reduce retear rates and improve short-term outcomes, although results vary by delivery method and tear severity (Yang et al., 2020). More recently, mesenchymal stem cell (MSC) therapies, particularly those using tenogenically primed subtypes from adipose tissue, have shown promise in improving collagen fibre organisation and mechanical strength in preclinical models (Chen et al., 2022b).

Amid these approaches, scaffold technologies have emerged as a promising strategy, particularly in cases where tissue gaps must be filled, or native architecture must be

replicated. Acting as a bioactive framework that mimic the body's natural extracellular matrix (ECM), a scaffold allows for essential cellular activities, such as attachment, proliferation, and differentiation, hence promoting new tissue growth and cell migration to for occupying the gaps in tears. Scaffolds derived from non-cross-linked human dermis ECM, have already begun to prove effective for rotator cuff repair, offering a compatible surface supporting collagen formation and tissue integration (Chen et al., 2022; Lu and Thomopoulos, 2013).

However, caution is advised with scaffold materials. Certain synthetic grafts and scaffold materials have triggered inflammatory responses or degraded prematurely, undermining their intended benefit (Hutmacher, 2000; Abdalla and Pendegrass, 2023). In addition, ECM of the enthesis is a complex continuous structure with gradual decrease in collagen fibre organization, aiding in stress distribution, load bearing, and acting as a barrier for cellular communication (Font Tellado, Balmayor and Van Griensven, 2015). This makes it challenging to replicate with grafts, resulting in poor tissue integration and limited long-term functionality (Chen et al., 2022).

Thus, research continues to refine scaffold materials and mimicking native structure, to maximize therapeutic potential while minimizing adverse responses.

3.5 Scaffold Fabrication Technologies

Various scaffold manufacturing technologies exist, each with unique advantages and limitations. These include:

- **Electrospinning (ES)**: A technique that uses an electric field to draw polymer solutions into fine fibres, creating scaffolds with high surface area and porosity (Byrne, 2021).
- **3D Bioprinting**: A method that deposits bioinks layer-by-layer to create complex tissue structures with precise control over cell placement (Mehrpouya et al., 2021).
- **Computer Aided Wet Spinning (CAWS)**: A technique utilising computer-controlled deposition into a coagulation bath to develop 3D scaffolds with a polymeric solution filament (Ozipek and Karakas, 2014; Mehrpouya et al., 2021).

- **Melt Electrowriting (MEW)**: An advanced technique that combines principles of electrospinning with precise polymer deposition at higher temperatures, allowing for the creation of highly organized scaffolds (Byrne, 2021).

MEW is the preferred choice for this investigation due to its advanced precision in controlling scaffold architecture. To create highly organized fibre structures that closely mimic the natural ECM is essential, and unlike other methods, MEW provides the necessary control over fibre alignment and spacing, critical for replicating the hierarchical structure of tendons. This level of scale with precision is not achievable with techniques like 3D bioprinting and ES, and though CAWS is a viable alternative with smaller possible diameters, MEW is solvent free, more efficient, better controlled, and additionally environmentally friendly, making it the superior choice (Bubakir et al., 2018).

The limitation to MEW is it requirement for the polymer to be heated, limiting the material selection to thermoplastics and risking degradation. However, for this project utilising PHA, and considering its advantages, this limitation can be overlooked.

3.6 Melt Electrowriting (MEW)

Melt electrowriting (MEW) is a versatile additive manufacturing technique that allows for the fabrication of scaffolds with high spatial resolution, as well as control over fibre orientation, diameter, and porosity. Unlike traditional electrospinning which uses a high-voltage electric field to draw polymer solutions into fine fibres, the original technique from which it was derived, MEW involves the precise electrical field-driven extrusion of polymer melts, mimicking closer to tradition AM. This process enables the creation of fibres with diameters ranging from several hundred nanometres to tens of micrometres in a controlled and arranged fashion (Brown, Dalton and Hutmacher, 2016).

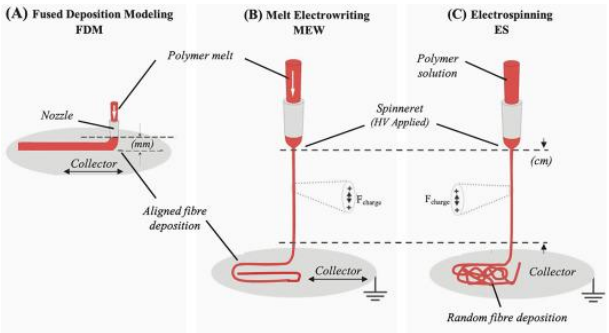

Figure 2: FDM vs. MEW vs. ES (V.F. Cardoso et al., 2022)

The key difference between them lies in the control over fibre deposition (Figure 2). While ES produces random non-woven mesh of fibres, MEW allows for precise control over fibre size, shape, and spacing (Bubakir et al., 2018). This level of control is particularly crucial for enthesis repair applications, where the alignment of the tendon's fibroblasts transitioning to mineralized type 1 collagen's spatially regulated structure, plays a vital role in mechanical properties and cellular behaviour (Lu and Thomopoulos, 2013).

Temperature control in MEW processing represents a critical parameter that significantly influences the quality of the final scaffold, influencing the polymer's viscosity and shear rate. The polymer melt must maintain consistent viscosity throughout the fabrication process to ensure uniform fibre formation. Applied voltage affects the electrostatic forces driving fibre formation and deposition, while collection speed determines fibre spacing and overall scaffold architecture (Brown, Dalton and Hutmacher, 2016). The interplay between these parameters requires careful optimization to achieve desired scaffold properties.

3.7 Material Selection and Properties

Table 1: Biomaterial comparison for ligament tissue engineering (Font Tellado, Balmayor and Van Griensven, 2015)

Material	Source	Biocompatibility	Bioactivity	Mechanical Strength	Degradation	Notes
MCL-PHA (Guo et al., 2022)	Microbial fermentation	High	High	Medium	Wholly degradable, depends on chain length, weeks to months	Low inflammation, bio-based
PCL (Dwivedi et al., 2020)	Synthetic (petrochemicals)	Moderate	Low	High	Slow, within years	Requires surface modification
Collagen (Caballé-Serrano et al., 2020)	Native ECM	High	High	Low	Fast, within days to weeks, controlled with elastin	Rapid resorption
Silk (Sobajo et al., 2008)	*Bombyx mori* (Silk moth)	Moderate	Moderate	High	Slow, controlled, within years	Needs pre-treatment
PGA (Mhiri et al., 2017)	Variety of methods	High	Low	High	Moderate, within months	Toxic acidic byproducts

As mentioned earlier, scaffold materials play a vital role in tendon tissue engineering, influencing everything from mechanical stability to cellular response and degradation rate. An ideal scaffold material should offer high biocompatibility, structural integrity, and support for cell adhesion and proliferation (Table 1). In this project, three materials stood out due to their relevance in literature and current clinical or experimental use: polycaprolactone (PCL), collagen, and polyhydroxyalkanoates (PHAs).

3.7.1 Polycaprolactone (PCL)

Polycaprolactone (PCL) is a synthetic biodegradable polymer widely used in tissue engineering due to its excellent mechanical properties and biocompatibility. It has a slow degradation rate, making it suitable for long-term applications (Arakawa and DeForest, 2017). However, PCL is derived from petrochemical sources, and although biocompatible, it has been associated with moderate inflammatory responses and poor native cell adhesion unless surface modifications are applied (G. Patricia Sailema-Palate et al., 2016; Elango, Zamora-Ledezma and Maté-Sánchez de Val, 2023).

3.7.2 Collagen

Collagen is a natural protein that forms the primary structural component of the extracellular matrix in various tissues. It is highly biocompatible and promotes cell adhesion, proliferation, and differentiation, making collagen-based scaffolds a strong candidate to mimic the natural extracellular matrix (Arakawa and DeForest, 2017). However, its mechanical strength and structural stability is lower than synthetic polymers under load, like PCL, can degrade more rapidly in vivo (Elango, Zamora-Ledezma and Maté-Sánchez de Val, 2023).

3.7.3 Polyhydroxyalkanoates (PHAs)

PHAs represent a family of microbially derived, bio-based polymers with significant potential in tissue engineering applications, offering an excellent balance between synthetic and natural material properties. They are biodegradable, biocompatible, able to support the growth of most tissue cells, and can be tailored by adjusting their monomer composition, making them flexible in a wide variety of applications. Unlike PCL, PHAs elicit low inflammatory responses, and their degradation rates can be controlled to match the healing process. Importantly, PHAs are non-toxic and

metabolised by natural cellular pathways, reducing risk during clinical translation (Guo et al., 2022). Three main subtypes exist:

- Short-chain length (SCL-PHA): stiff, brittle, high-crystallinity, thermoplastic behaviour.
- Medium-chain length (MCL-PHA): lower crystallinity, soft, elastomeric, and more suitable for soft tissue scaffolds.
- Long-chain length (LCL-PHA): highly elastic with very low crystallinity (Basnett, Ravi and Roy, 2016; Mai et al., 2024).

3.7.4 Advantages of PHA

While PCL is a well-established MEW-compatible material and collagen closely mimics the native ECM, both present significant drawbacks. PCL's slow degradation and synthetic origin raise biocompatibility concerns, particularly in inflammatory environments. Collagen, though biologically optimal, lacks the mechanical durability needed for load-bearing applications like tendon repair and degrades too rapidly for scaffold-guided regeneration.

MCL-PHA, on the other hand, offers a compromise between mechanical performance and biological function. Its flexibility enables elastic deformation without fracturing, and its predictable degradation profile allows for better alignment with tissue regeneration rates. Additionally, being bio-sourced and metabolically safe, MCL-PHA represents a sustainable and translationally relevant material — particularly when scaffold degradation needs to align with biological healing timelines.

Moreover, PHAs can be combined with bioactive molecules to enhance their regenerative potential. For example, PHAs can be loaded with growth factors like fibroblast growth factor (FGF) to promote tendon healing and regeneration (Guo et al., 2020).

4. Aims and Objectives

The overarching objective of this project is to develop and optimize a melt electrowriting (MEW) protocol for medium-chain length polyhydroxyalkanoate (MCL-PHA). With this established, it opens further study into fabricating scaffolds suitable for tendon repair, targeting the enthesis region of the rotator cuff. The aim is to produce thin, aligned, and structurally stable fibre architecture with a consistent printing process. This is crucial for further down the line when developing unique architecture and patterns that enable oriented cell growth by mimicking native tissue organization.

To achieve this, the project is divided into three experimental phases, each with specific goals:

4.1 Phase 1 – MEW Parameter Benchmarking with PCL

The first phase establishes a benchmark using PCL, a well-characterized and MEW-compatible material. This phase builds the foundational knowledge necessary to approach PHA, providing a reference for material-specific optimization. The goal is to:

a) Understand the functional dynamics of MEW and how each parameter — voltage, pressure, temperature, collector distance, and speed — affects print quality.

b) Identify a set of optimal printing parameters for PCL using both trial-and-error and Design of Experiments (DOE) methodologies.

c) Use light microscopy for structural characterization, evaluating fibre uniformity, accuracy, and print process stability.

4.2 Phase 2 – Parameter Optimization for PHA

The second phase transitions to printing with PHA. This phase focuses on understanding and overcoming the material-specific challenges of MCL-PHA in the MEW context. The goals here are to:

a) Systematically adjust and optimize MEW parameters to achieve thin, consistent, non-merging fibres and accurate multi-layer prints. In addition, the printing process needs to remain consistent and stable to ensure repeatability and avoid defects.

b) Observe the printing process via light microscopy and analyse structure using light microscopy and scanning electron microscopy (SEM). Light microscopy will provide an overview of the scaffold structure, while SEM will offer detailed insights into fibre morphology and surface characteristics, which are crucial for understanding cell-material interactions, and more importantly, fibre stacking.

c) Analyse microscopy images to quantitatively measure fibre diameters, improving the accuracy and objectivity of characterisation. Using these diameter values, apply a refined DOE to narrow parameter spaces.

In recognition of the fact that conventional tools such as ImageJ and Fiji proved insufficient in reliably detecting and measuring fibres from microscope images, a new sub-objective is added:

d) Develop a custom image analysis program for automated fibre diameter measurement after recognising the limitations of existing established software ImageJ. A bespoke Python-based solution was created to process microscopy images, calibrate spatial measurements, and accurately quantify fibre dimensions.

This software became essential for generating quantitative data needed for DOE analysis and for validating the consistency of optimised print settings.

4.3 Phase 3 – Scaffold Design and Validation

With optimal parameters identified, the final phase involves testing their robustness in more complex scaffold configurations. The development of scaffold designs draws from detailed analysis of native tendon architecture and mechanical behaviour. Recent advances in understanding tendon tissue mechanics have highlighted the importance of the hierarchical structure in determining functional properties (Martinez et al., 2023). This phase serves as a proof-of-concept for translating optimized MEW parameters into functional, biomimetic scaffold designs tailored for tendon regeneration. The goals are to:

a) Develop custom scaffold patterns that incorporate variable grid densities, mimicking the structural gradients observed in native enthesis tissue. Through

this, we can determine if it is possible to achieve different grid densities in a single print.

b) Test the structural integrity and layering accuracy of these patterns using the previously optimized MEW settings, and evaluate repeatability and scalability of the process, including prints extending to thirty layers.

5. Research Materials and Methods

Table 2: Experimental protocols

Component	Details
Bioprinter	RegenHU 3D Discovery, equipped with Melt Electrowriting (MEW) printhead kit
Collector	Glass petri dishes
Ambient Control	Printing room temperature regulated
Material – PCL	Pellets heated to 100 °C for 1 hour at startup
Material – PHA	Heated to 55 °C for 45 minutes for standard runs; final trials used 100 °C for 30 minutes startup and maintained at that temperature throughout
Needle Gauge	26G (0.25 mm inner diameter)
Scaffold Dimensions	30 mm × 10 mm rectangular scaffold; 0.5 mm fibre spacing; 0.001 mm layer height; square grid pattern with 90° orientation
Layer Counts	Standard prints: 5-10 layers; robustness testing: 30 layers

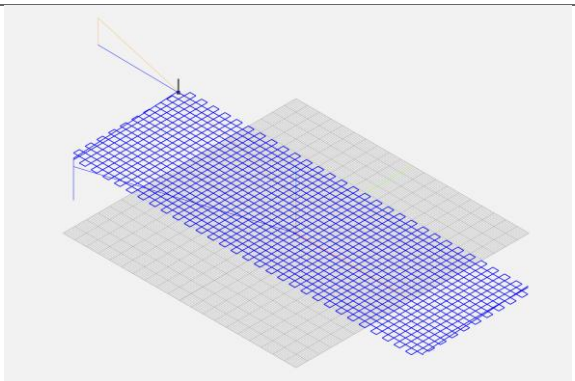

Figure 3: Standard scaffold design - preview through NC Viewer

All experimental work was conducted using the RegenHU 3D Discovery multi-head bioprinter equipped with a melt electrowriting (MEW) printhead kit. Fibres were printed onto glass petri dishes cleaned with 70% v/v ethanol between every run, and ambient

temperature was regulated to avoid overheating the print and irregularities with prolonged use. For PCL, pellets were inserted and maintained at 100 °C for 1 hour, ensuring it was fully melted, and all air bubbles were released. Similarly, for PHA, the polymer was inserted and maintained at 55 °C for 45 minutes, though for the last set of trial runs, the polymer was kept at the maximum temperature setting of 100 °C for 30 mins for startup and maintained at that temperature throughout the process.

A 26G (Ø 0.25 mm I.D.) was used for all documented runs to fabricate scaffolds of 5 to 10 layers, with 0.001 mm layer height, 30 mm length and 10 mm width, with a 90° angled design of square grids with a fibre separation distance of 0.5 mm (Figure 3). To evaluate the robustness of certain settings, repeatability tests are run for 30-layer grids to observe if the printing process remains consistent over longer runtimes (Table 2).

The printer was operated via BioCAD, a proprietary software package supplied by RegenHU. BioCAD provided the interface for designing print patterns, controlling deposition parameters, and generating the G-code-like .iso files required for executing prints.

A significant advantage of the RegenHU setup was its integration with a dedicated air compressor system, enabling the application of higher and more stable pneumatic pressures compared to standard laboratory air supplies. This enhanced the controllability of the extrusion pressure parameter during MEW, allowing a broader range of processing conditions to be explored.

In the entirety of this project, four main constants were: the printer hardware (RegenHU 3D Discovery), ambient temperature, the needle size, and the materials used. Polycaprolactone (PCL) was selected for initial benchmarking of MEW parameters, while medium-chain length polyhydroxyalkanoate (MCL-PHA) was the target material for parameter optimisation and scaffold development.

6. Results and Discussion

6.1 Phase 1 – Initial Familiarisation with MEW and PCL

6.1.1 Results

(Figure 4) displays the comparison between an initial trial print and the final benchmark print under an analogue light microscope with a 10x objective zoom. (Table 3) are the parameter settings of the benchmark print, printed up to 10 layers.

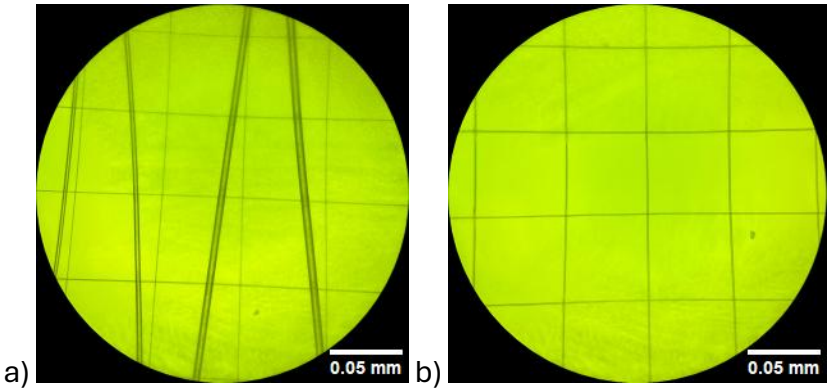

Figure 4: PCL light microscopy images. a) Initial trial print, b) Final benchmark print

Table 3: Benchmark PCL settings

Temperature (°C)	Pressure (MPa)	Distance (mm)	Voltage (kV)	Speed (mm/s)
80	0.3	4	6	15

6.1.2 Discussion

The experimental phase began with understanding the operational intricacies of the RegenHU 3D printer. Given the high sensitivity of melt electrowriting to parameter variations, the priority was to establish stable printing with a well-documented material: PCL. As a well-used polymer in MEW research, PCL provided a reliable basis for developing process intuition and identifying early success criteria (Shahverdi et al., 2022).

Parameter	Set 1	Set 2	Set 3
Distance [mm]	1	3	6
Feed Rate [mm/s]	5	15	25
Temperature [C]	80	80	80
Voltage [kV]	2.5	6	10
Pressure [MPa]	0.2	0.3	0.3

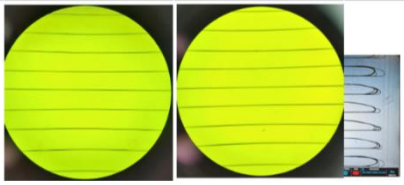

Figure 5: Baseline parameter boundaries from initial trials for establishing DOE

Initial prints were conducted via trial and error, using random default settings as a baseline. This early stage was essential in understanding the qualitative cues associated with a stable print: formation of a Taylor cone, minimal pulsing, consistent fibre thickness and shape, and fibres stacking cleanly into layers. These cues became important later when evaluating PHA and its unknown behaviour, forming the six true/false factors to rank the printing process.

Table 4: Generated values through JMP. Refer to Appendix C for full tables with diameter and overall rank.

Σ	Pattern	Temperature (C)	Pressure (MPa)	Distance (mm)	Voltage (kV)	Feed Rate (mm/s)	Diameter	Extrusion	Consistent Jet	Appropriate Lag	No fibre jumping	No fibre pulsing	No merging
1	−000−	75	0.25	3.5	7	5	•	•	•	•	•	•	•
2	− −000	75	0.1	3.5	7	17.5	•	•	•	•	•	•	•
3	−0+00	75	0.25	6	7	17.5	•	•	•	•	•	•	•
4	−0−00	75	0.25	1	7	17.5	•	•	•	•	•	•	•
5	−00+0	75	0.25	3.5	12	17.5	•	•	•	•	•	•	•
6	−00−0	75	0.25	3.5	2	17.5	•	•	•	•	•	•	•
7	− +000	75	0.4	3.5	7	17.5	•	•	•	•	•	•	•
8	−000+	75	0.25	3.5	7	30	•	•	•	•	•	•	•
9	0−00−	80	0.1	3.5	7	5	•	•	•	•	•	•	•
10	00−0−	80	0.25	1	7	5	•	•	•	•	•	•	•
11	00+0−	80	0.25	6	7	5	•	•	•	•	•	•	•
12	000− −	80	0.25	3.5	2	5	•	•	•	•	•	•	•
13	000+ −	80	0.25	3.5	12	5	•	•	•	•	•	•	•
14	0+00−	80	0.4	3.5	7	5	•	•	•	•	•	•	•
15	0−0−0	80	0.1	3.5	2	17.5	•	•	•	•	•	•	•

Following, a structured Design of Experiments (DOE) was initiated by using the software JMP (Pro 18) to map how five independent variables — voltage, collector distance, printhead speed (also known as scanning speed), pressure, and melt temperature — affected printing outcomes. The parameter ranges were based on known PCL ranges from initial trials (Figure 5), and the DOE design generated experimental runs used a Box-Behnken model, most representing a unique parameter set within the defined constraints, with some repeated to test variability from repeating runs (Table 4).

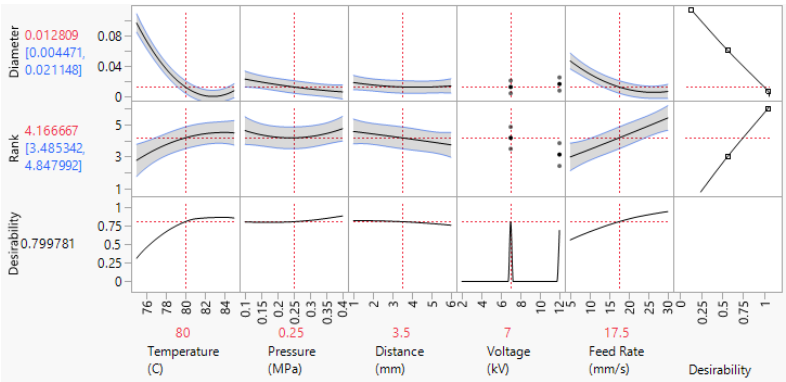

Figure 6: Generated profiler through JMP; based on standard least squares model

Simply running all experiments from the DOE yielded critical insights, with one of the runs helping converge on a configuration that consistently produced clean, thin fibres across multiple layers (Figure 4). This setting served as the benchmark for later phases, completing phase 1 objective (a) and (c) (Table 3). However, it was during this exercise that a key limitation surfaced: there was no quantitative and automated method for evaluating fibre diameters. Visual observation alone was insufficient for statistical or comparative validation, especially with five variables at play.

This realization prompted the decision to develop an image analysis tool, which would later become central to the entire optimisation workflow. After developing the measurement software, the PCL DOE was modified and completed, generating the profiler and developing the parameter relationship to diameter and rank (Figure 6). Upon maximising desirability (minimising diameter while maximising rank), the optimised parameters were remarkably close to the benchmark settings, reinforcing its validity and completing phase 1 objective (b).

6.2 Phase 2 – Transitioning to and Troubleshooting PHA Printing

6.2.1 Results

(Table 5) Displays the final low temperature parameter settings optimised manually through trial and error that best match the goal of thin fibres. However, it does not print accurately due to high voltage causing fibre jumping, pulsing that breaks the Taylor cone jet, resulting in an unstable printing process. (Figure 7 a) displays the initial trial prints with PHA using the same printing parameters as PCL except for a lower temperature of 50 °C, for 5 layers. In contrast, (Figure 7 b) is the parameter settings mentioned earlier, for

10 layers also. (Figure 7 c) is the same low temperature manually optimised parameter settings for 30 layers. Microscope images were taken under a digital microscope at 4x objective, saved as TIFF files for high resolution.

Table 5: Final trial and error low temperature parameter setting.

Temperature (°C)	Pressure (MPa)	Distance (mm)	Voltage (kV)	Speed (mm/s)	Ambient (°C)
52	0.015	4	5	15	23.3

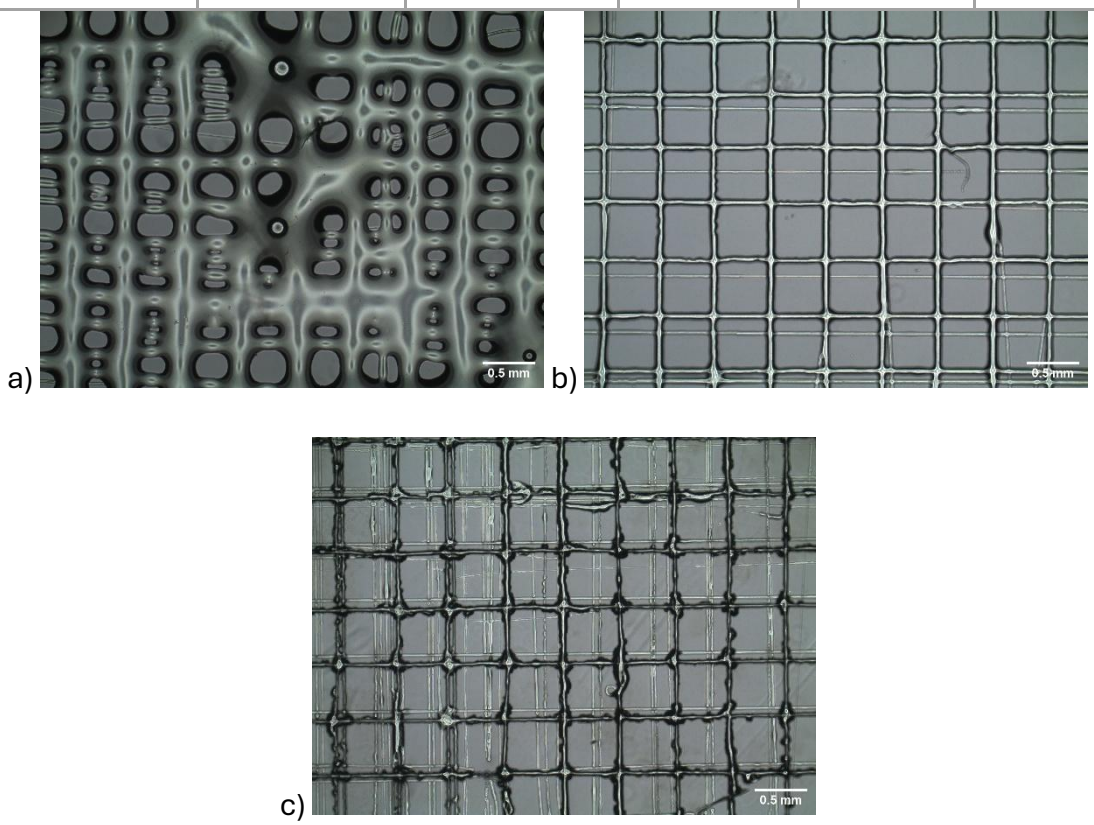

Figure 7: PHA manual optimised light microscopy images. a) Initial trial print, b) Final low temperature print, c) 30-layer low temperature print

6.2.2 Discussion

With PCL optimisation complete, the focus shifted to medium chain length PHA. Despite its favourable biological properties and biodegradable profile, PHA presented notable challenges in MEW. The material exhibits low crystallinity, unlike the semi-crystalline PCL, low viscous behaviour at lower temperatures due to high shear rates, making it slow to cool and crystallize, causing layers to merge rather than stack (Figure 8).

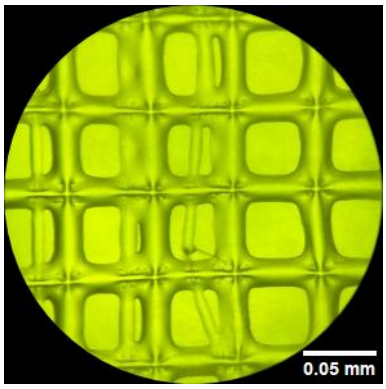

Figure 8: Initial PHA Prints

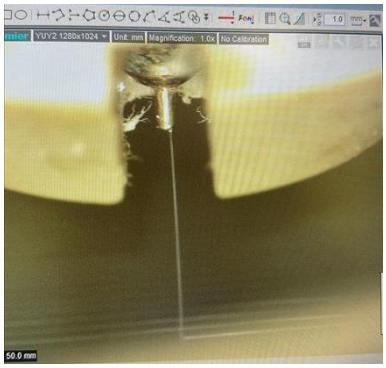

Figure 9: High voltage printing process through camera

Initial prints with PHA were unsuccessful (Figure 7 a). Fibres appeared thick and irregular; layers spread outwards rather than building vertically. Extensive trial and error followed, aiming at identifying which variables most influenced fibre thickness and jet stability. Voltage, temperature, and pressure were of particular concern. Several settings that worked well for PCL caused no lag, immediate pulsing, or even spraying when applied to PHA (Figure 9).

What became apparent was that cooling dynamics were central to the failure of multi-layered prints. Even using a refrigerator or regulating the ambient temperature failed to solve the issue completely; the material simply would not permit fast solidification unless cooled naturally at room temperature. Interestingly, the PHA remained a viscous liquid after thawing from cold (2.5 – 4.5 °C) and freezing (-21 °C) conditions, an observation that reinforced the need for higher process control rather than relying on environmental cooling alone.

Nevertheless, through parameter isolation and adjustment, a stable print setting was eventually reached (Table 5). Although the fibres were thin and the structure was coherent, the print process was only repeatable for up to five layers before defects appeared. The voltage was less than optimal, resulting in inconsistent fibre shape, and certain layers repelling off to the side (Figure 7 b).

To further evaluate the print's integrity, the pattern was printed up to thirty layers to assess the extent of defects and resulting print quality. As observed earlier, around the 5-to-6-layer mark, the extrusion jet broke due to continuous build up at the nozzle, and it entered a cycle of attempting to stabilize a Taylor cone and a pulse breaking it once more. The result observed under the microscope was inaccurate, misaligned, and inconsistent in shape throughout the build height (Figure 7 c).

6.3 Development of Custom Fibre Measurement Software

To move beyond visual evaluation, a custom Python-based software tool was developed to measure fibre diameters from microscopy images, completing phase 2 objective (d). Established tools like ImageJ/Fiji failed to provide consistent or accurate readings due to the transparency of fibres, leading to poor edge detection and measurements across variable image qualities.

The custom program begins with a calibration step: an image of a ruler under the same microscope and zoom/objective as the fibre images is analysed to calculate the pixel-to-millimetre ratio. This value defines the spatial resolution of each subsequent image.

Each fibre image is then processed through a multi-step pipeline (Figure 10): first, grayscale conversion and normalization, then the application of Gaussian blur, followed by adaptive thresholding to binarize the image. Canny edge detection and Hough line transformation are applied onto the mask, and finally, cross-sectional intensity profiling for each line and some statistical filtering.

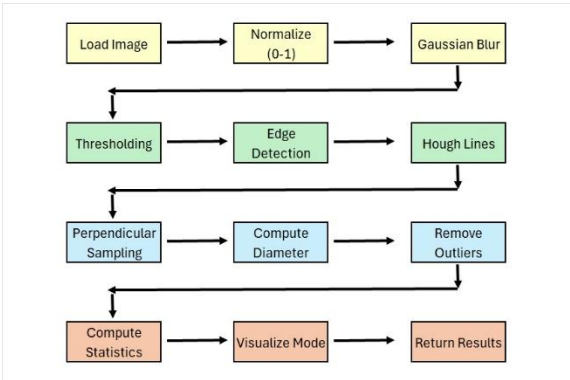

Figure 10: Software Process Pipeline

6.3.1 Calibration and Grayscale Conversion

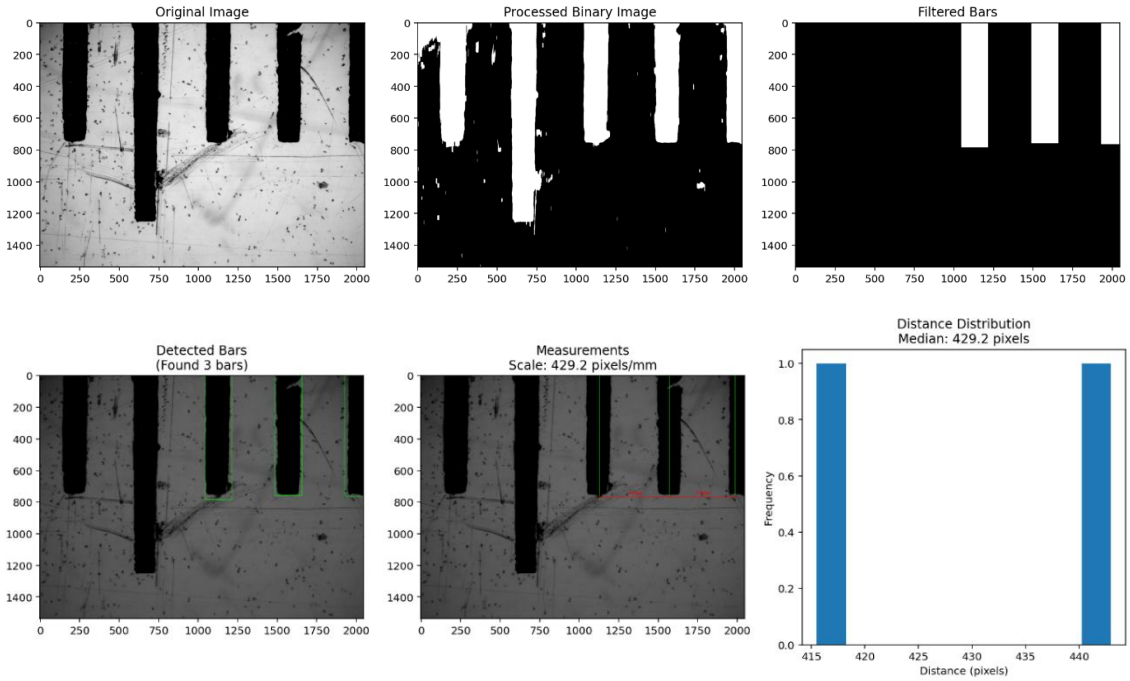

Figure 11: Example of Visualised Overlay of Calibration

The calibration step is performed using an image of a standard ruler under the same microscope objective as the scaffold images (Figure 11). The distance between known tick marks was measured in pixels, establishing a pixels-to-millimetre conversion factor:

$$Scale\ Factor\ (px/mm) = \frac{Pixel\ Distance}{Known\ Distance}$$

Each scaffold image was first converted from RGB to grayscale, mapping the three colour channels into a single intensity channel. Grayscale conversion reduces image dimensionality, improves processing speed, and focuses analysis purely on structural features rather than colour information.

6.3.2 Gaussian Blurring and Binary Mask Generation

To reduce high-frequency noise that could interfere with edge detection, a Gaussian blur was applied to the grayscale image. Gaussian blurring smooths pixel intensity variations by convolving the image with a Gaussian function:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

Where σ represents the standard deviation controlling the blur intensity. Blurring makes fibre structures more coherent while suppressing small-scale noise, ensuring that the edges detected later correspond to real fibre boundaries rather than imaging artefacts. On top of this, an adaptive threshold algorithm was used to convert the blurred grayscale image into a binary mask. This method computes thresholds locally, adapting to varying illumination across the image. Pixels are classified as either background (0) or foreground (1), producing clear separation between fibres and background noise (Figure 12).

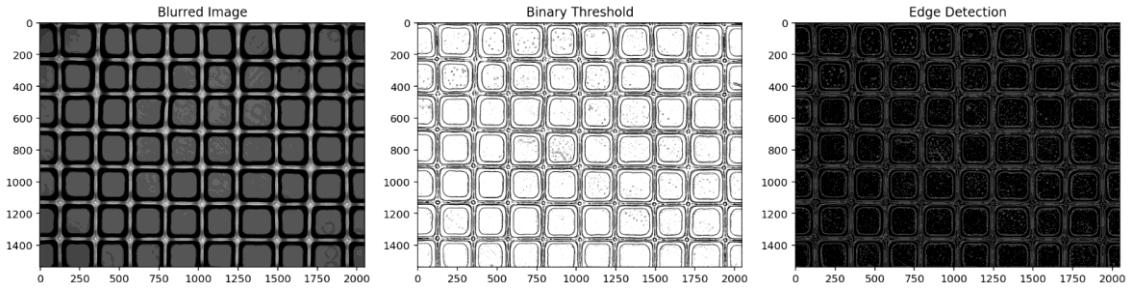

Figure 12: Example of image processing at each stage: gaussian blur, binary mask, and canny edge detection

6.3.3 Edge Detection and Cross Section Analysis

To accurately identify fibre boundaries, Canny edge detection was applied to the binary image (Figure 12). It is a complex algorithm that produces a set of crisp, connected edges corresponding to fibre outlines. Following edge detection, the Hough Transform was used to detect straight line segments across the fibre images. Each line corresponds to a potential cross-section suitable for diameter measurement.

For each detected line, a 1D intensity profile was extracted, spanning perpendicular to the fibre axis. The fibre diameter was determined by locating the first and last threshold crossings in the intensity profile — the points where intensity values crossed a set midpoint between fibre and background intensities.

6.3.4 Statistical Filtering and Diameter Calculation

With an intensive list of raw diameter measurements containing noise and outliers, particularly from poorly segmented regions, a statistical filtering process was applied to improve measurement reliability. The approach was based on Z-scores, where the Z score of each diameter value was calculated:

$$Z_i = \frac{d_i - \mu}{\sigma}$$

Where μ is the mean of all measured diameters. Z-scores quantify how many standard deviations a data point lies from the mean. Measurements with an absolute Z-score less than 3 were retained, corresponding to diameters within three standard deviations of the mean. This threshold effectively captures approximately 99.7% of data in a normally distributed population, removing extreme outliers without overly restricting valid variation. Final fibre diameter was calculated using the scale factor:

$$D_i\ (mm) = d_i\ (px) \times Scale\ Factor(px/mm)$$

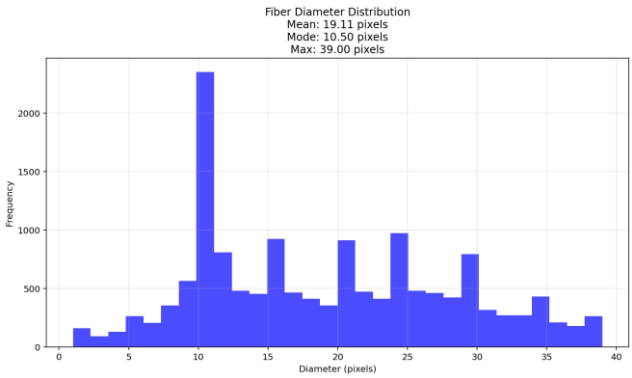

Figure 13: Example of histogram of processed diameter results.

The program provides both raw measurements and visual overlays showing which fibres were measured and where, along with a histogram displaying the frequency of diameters, with the mode being the assumed fibre diameter of the print (Figure 13). These outputs dramatically improved the reliability of DOE analysis by adding quantitative precision to qualitative assessments.

While it is limited by the resolution of the 4x objective lens (~2.33 μm/pixel), it was sufficient for identifying trends and assessing print consistency. High-magnification versions can be considered for future iterations of the software.

6.4 Final DOE for PHA and Discovery of High-Temperature Breakthrough

6.4.1 Results

(Table 6) is the optimised low temperature parameter settings optimised through DOE for pressure and voltage, displayed by (Figure 14 a) for 10 layers. (Figure 14 b) is the initial

high temperature print conducted. After optimisation, final high temperature parameter settings were established (Table 7), resulting in (Figure 14 c). Microscope images were taken under a digital microscope at 4x objective, saved as TIFF files for high resolution.

Table 6: Final low temperature optimised parameter settings through DOE

Temperature (°C)	Pressure (MPa)	Distance (mm)	Voltage (kV)	Speed (mm/s)	Ambient (°C)
50	0.1	3	7	15	23.2

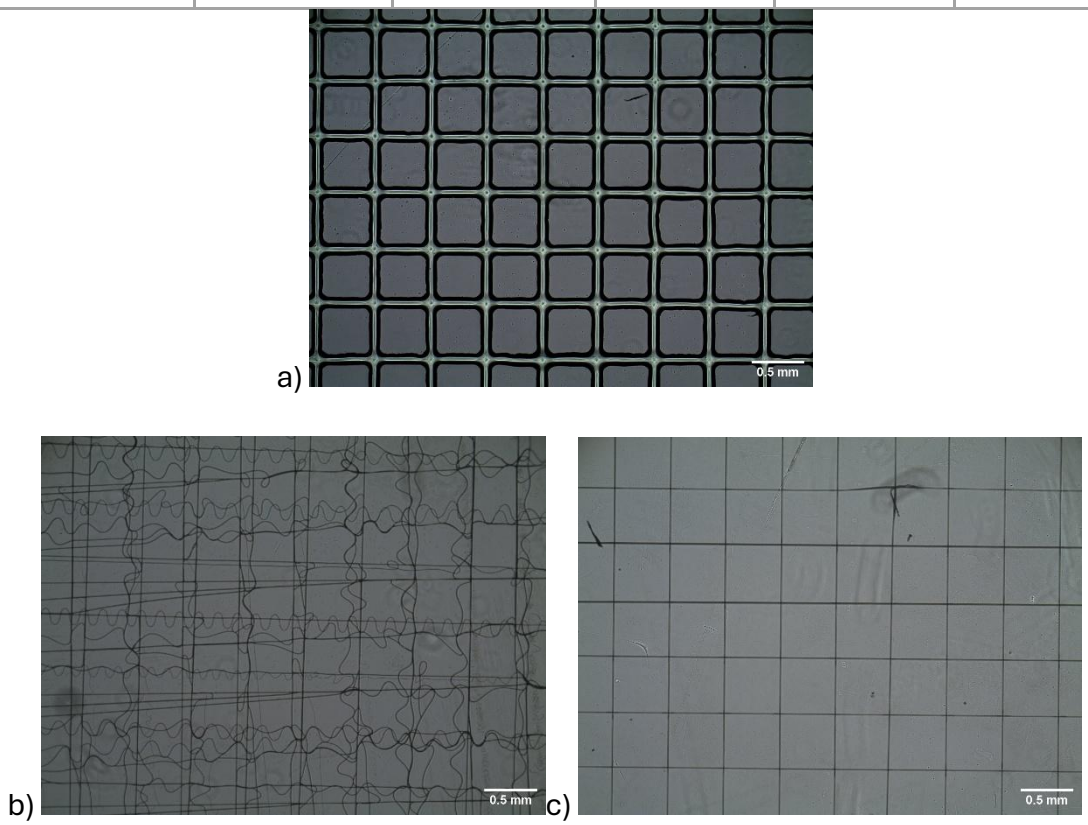

Figure 14: Final PHA light microscopy images. a) DOE optimised low temperature print, b) Initial high temperature trial, c) Optimised high temperature print

Table 7: Final high temperature parameter settings

Temperature (°C)	Pressure (MPa)	Distance (mm)	Voltage (kV)	Speed (mm/s)	Ambient (°C)
100	0.1	2.5	4	20	23.4

6.4.2 Discussion

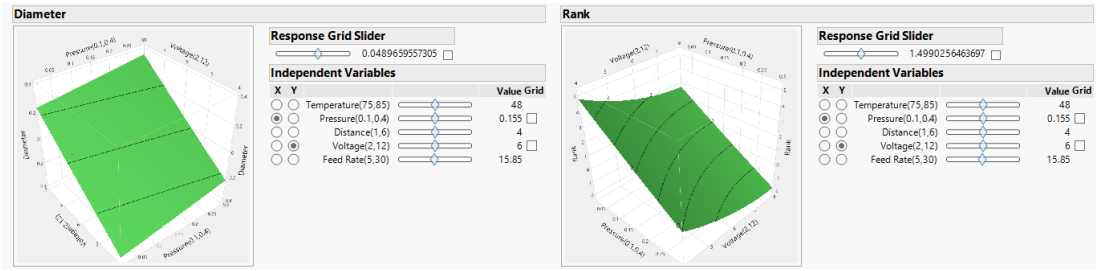

Figure 15: Generated Surface Plots for Pressure vs. Voltage based on Diameter and Rank

With the software operational and qualitative behaviour of PHA better understood, a second DOE was conducted for PHA specifically, completing phase 2 objective (c). Parameter limits were adjusted based on previous observations through trial and error, narrowing the experimental space to pressure and voltage (Figure 15). Using this narrowed scope, a new series of prints were produced and analysed for fibre diameter, layering ability, and process reliability. As per the DOE, maintaining parameters around atmospheric pressure (0.1 MPa) and 7 kV voltage would result in high rank and low diameter prints.

Upon further investigation, the DOE succeeded in producing thinner fibres and better print quality than through trial and error (Table 6), completing phase 2 objective (a). However, a major limitation persisted – all runs were accompanied by minimal pulsing, and no combination of pressure, speed, or voltage could overcome the material merging each layer (Figure 14 a).

Towards the end of the experimental phase, a paper (Marcello et al., 2024) processing similar MCL-PHA via 3D extrusion printing, explored temperatures up to 110 °C — far beyond the polymer's nominal melting point (40–60 °C). Despite concerns about potential thermal degradation, these conditions were trialled for melt electrowriting, mixing established parameters with elevated temperatures: 100 °C, 0.1 MPa, 15 mm/s, 5 kV, and 3 mm collector distance as starting values.

The result was surprising and transformative. Observed fibres were thin, extrusion was continuous, and highly stable, with a consistent printing process and no signs of pulsing or merging. More importantly, the material demonstrated vertical layer formation, suggesting faster solidification (Figure 14 b).

Through minor refinements based on the new thermal window, a new best-in-class setting was achieved, capable of clean, layered, and repeatable scaffold printing (Figure 14 c) — with the caveat that extended use, of approximately 10 to 14 hours, led to visible signs of PHA degradation through discoloration, and change in material properties as repeating parameter settings gave different print quality results. This necessitates periodic flushing of the material between runs.

6.5 Phase 3 – Custom Pattern Design and Robustness Testing

6.5.1 Results

(Figure 16Figure 16: Horizontal Gradient Print Example) shows a print using the modified g-code for a horizontal gradients, up to 5 layers. Microscope images were taken under a digital microscope at 4x objective, saved as TIFF files for high resolution. (Figure 17) are the Scanning Electron Microscope (SEM) images taken for the high temperature optimised parameter settings for 30 layers.

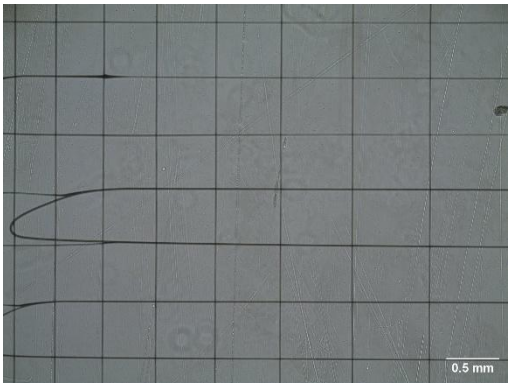

Figure 16: Horizontal Gradient Print Example

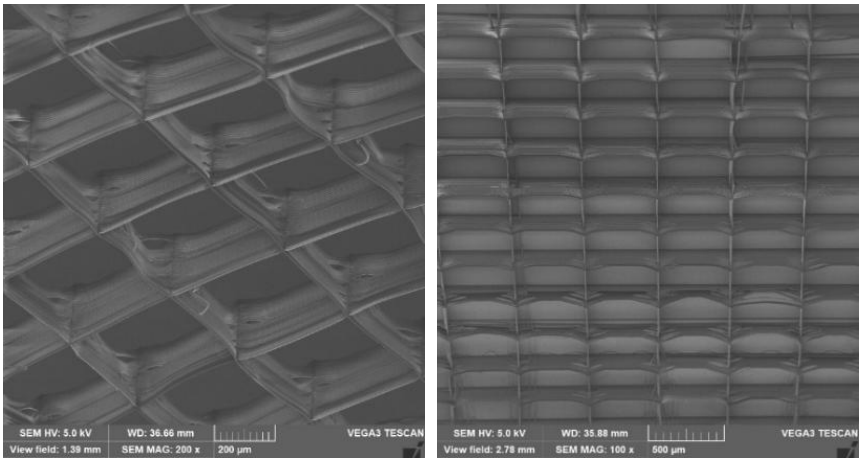

Figure 17: SEM Images of High Temperature 30 Layer Prints[1]

6.5.2 Discussion

With the optimized high-temperature MEW parameters established for PHA (Table 7), the final stage of experimental development aimed to test overall robustness under structurally complex conditions. Until this point, all tests had been performed on a uniform grid pattern.

Due to limitations in the proprietary BioCAD software, which made pattern creation complex and inflexible, a workaround was identified. By examining the exported .iso files, which serve as the printhead movement instructions for the RegenHU printer, through an online G-code editor NC Viewer, it became clear that manual editing of G-code could enable fully customised patterns without software constraints. This allowed for exploration beyond regular uniform grids into gradient meshes — structures with varying grid densities from one edge to another.

[1] SEM Images are inverted from negative tilt angles

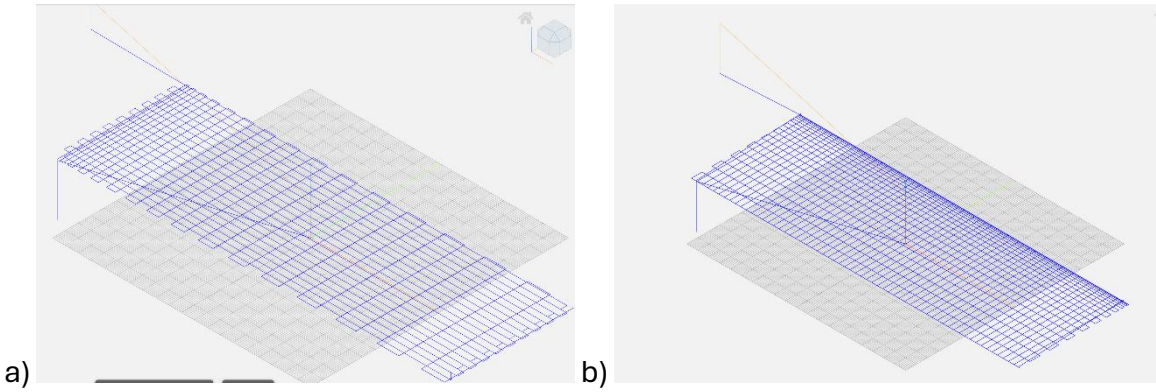

Figure 18: Custom G-Code preview through NC Viewer. a) Horizontal Gradient, b) Vertical gradient

Custom toolpaths were created (Figure 18) to produce vertical density gradients, where the spacing between horizontal lines increased progressively from top to bottom, and horizontal density gradients, where vertical spacing increased from left to right. This marks the completion of phase 3 objective (a).

These patterns served two purposes. First, they tested whether the printing process remained stable as fibre proximity changed, which can theoretically be impacted by local charge effects and instabilities in MEW. Secondly, it mimicked the gradient and material density of the enthesis region in a basic fashion, serving as an example for more complex mimicry patterns of the enthesis.

Each custom pattern required unique toolpaths to accommodate MEW's inherent constraint: the jet must remain continuous during deposition. This meant that toolpaths had to be carefully sequenced to prevent interruptions that would travel across prints or collapse the Taylor cone. Start-up sequences were maintained, and transitions between paths were minimised to preserve print flow.

Both gradient patterns were successfully printed without any major defects (Figure 16) – a strong indication that the optimised parameters held across varying fibre densities and toolpath geometries. The fibre diameters remained consistent throughout, and no pulsing or merging occurred, even in the densest regions.

To further evaluate structural integrity, the patterns were printed up to thirty layers as well, in accordance with phase 3 objective (b). In stark contrast to low temperature runs, the prints remained clean, aligned, and vertically consistent throughout the build height, with SEM images proving its consistent accuracy in deposition and layering (Figure 17). This

was a milestone result, confirming that the parameter set was but robust under varying scaffold patterns and geometries.

6.6 Practical Adjustments and Final Observations

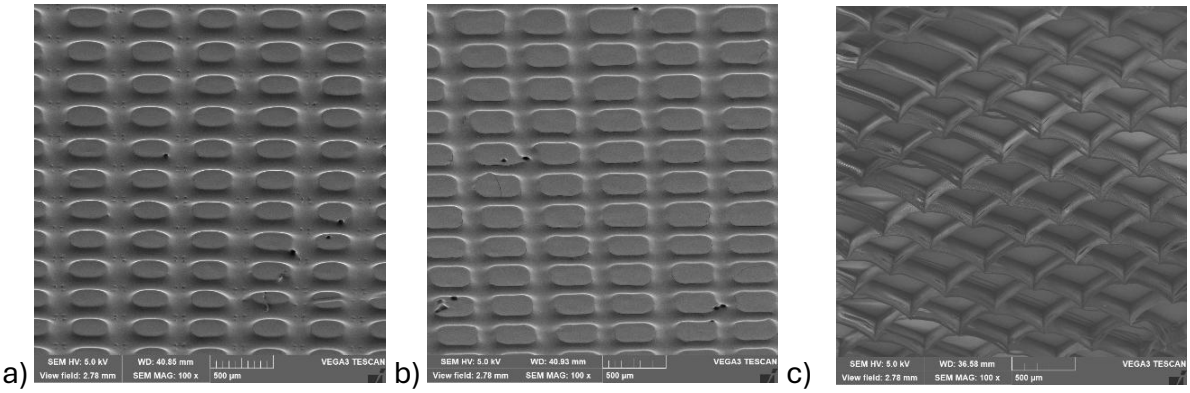

Figure 19: SEM images. a) Final low temperature trial and error parameters, b) DOE low temperature optimised parameters, c) High temperature parameters 30 Layers

Although the print settings themselves proved stable, minor machine-related adjustments were required to ensure print consistency at high speeds. Double-sided tape was used to secure the petri dish, eliminating movement entirely and ensuring layer alignment, and a custom 3D printed holder was used to position to position the dish at the centre. This highlights the importance of controlling external effects to achieve full command over these controlled and precise builds.

The project also relied on consistent behaviour of polyhydroxyalkanoate (PHA) batches. Slight variations in viscosity were observed between material preparations, and to manage this, process parameters were kept consistent across comparative tests, and any abnormal prints were re-evaluated to rule out material degradation or other random factors.

Further studies may involve other fabrication technologies in lieu of MEW, which could lead to different results and outcomes. A limitation to MEW is it requirement for the polymer to be heated, limiting the material selection to thermoplastics and risking degradation.

Finally, the best samples from each phase of PHA optimisation were submitted for scanning electron microscopy (SEM) imaging, in accordance with phase 2 objective (b). These analyses visually confirmed what the images and quantitative and qualitative data

had suggested throughout: only the final high-temperature prints demonstrated all the key traits of a successful MEW scaffold – thin, aligned, layered fibres with minimal merging and consistent morphology (Figure 19).

7. Conclusion and Future Directions

The primary objective of this project was to develop and optimise melt electrowriting (MEW) parameters for printing polyhydroxyalkanoate (PHA) scaffolds, targeting their application in rotator cuff tendon repair. By leveraging the regenerative potential of PHA and the architectural precision of MEW, the aim was to produce fibre-aligned scaffolds capable of mimicking the cell structure of the enthesis region.

This goal was approached in three phases: benchmarking the MEW process using PCL, adapting the workflow for PHA through extensive experimentation and design of experiments (DOE), and finally, testing the robustness of the optimised parameters on custom scaffold designs. The development of a bespoke fibre diameter measurement tool allowed the shift from subjective visual inspection to quantitative evaluation, improving the rigour of optimisation. The project culminated in the discovery of two results: the first is the optimised parameters discovered through trial and error and DOE, though the print itself does not achieve all that we intended. The second is a high-temperature print parameter settings, previously avoided due to valid concerns over polymer degradation, though with significantly enhanced print quality, fibre consistency, and layer-by-layer stacking.

However, several challenges and limitations remain. The resolution of the fibre measurement system, currently bound by the 4× microscope objective, limits to approximately 2.33 µm per pixel. This restricts the ability to distinguish subtle variations in fibre diameter, particularly for higher-definition prints where precise morphological comparison is necessary. Future work should address this by improving the program software to incorporate higher magnification optics to allow more detailed post-print analysis.

Another critical limitation is that while a DOE was performed for standard temperature ranges, no structured DOE has yet been conducted at the high-temperature settings that proved most effective. A systematic investigation into the influence of pressure, speed, and voltage under these new thermal conditions would help establish an even more robust and transferable parameter set.

Moreover, while the project succeeded in demonstrating structural fidelity and stability in simple and gradient grid designs, no formal quantitative analysis has yet been performed comparing the actual printed scaffold to the original G-code. Assessing geometric fidelity — including fibre-to-fibre spacing, alignment, and pattern accuracy — is essential for validating the translational potential of the process.

Finally, although custom G-code was manually created for gradient mesh designs, the approach remains basic. More advanced algorithms capable of generating anatomically informed, spatially graded patterns that would better mimic the heterogeneous fibre arrangements of the native enthesis can provide more information on real-world applications. This would require both CAD and G-code generation strategies that incorporate biological design rules, informed by structure of the native tissue.

As research into PHAs expands, their potential for use in advanced, tissue-specific scaffold design is becoming more apparent. Continued development of custom polymer blends, functional surface coatings, and tailored chain-length variants could further enhance their suitability for applications such as rotator cuff repair (Mehrpouya et al., 2021). Moreover, combining PHA-based scaffolds with bioactive molecules or cells may accelerate healing and integration, pushing the material beyond simply structural support into active therapeutic roles (Basnett, Ravi and Roy, 2016).

Future research should also focus on performing rheology on the scaffold, to determine their mechanical properties, and the long-term in vivo performance of PHA-based scaffolds, including their integration with host tissues and their ability to support functional tendon regeneration. Future work can also involve other fabrication technologies instead of MEW, which would require a different approach to optimisation, leading to different results and outcomes. However, this paper provides a logical and constructive overview on how to tackle optimisation and can be utilised as reference.

In conclusion, this project successfully establishes a reproducible MEW protocol for PHA scaffolds with significantly improved print quality, structural fidelity, and pattern control. While current outcomes meet the objectives defined at the outset, several areas for technical refinement and deeper quantitative validation remain and addressing these will further enhance the application of the work.

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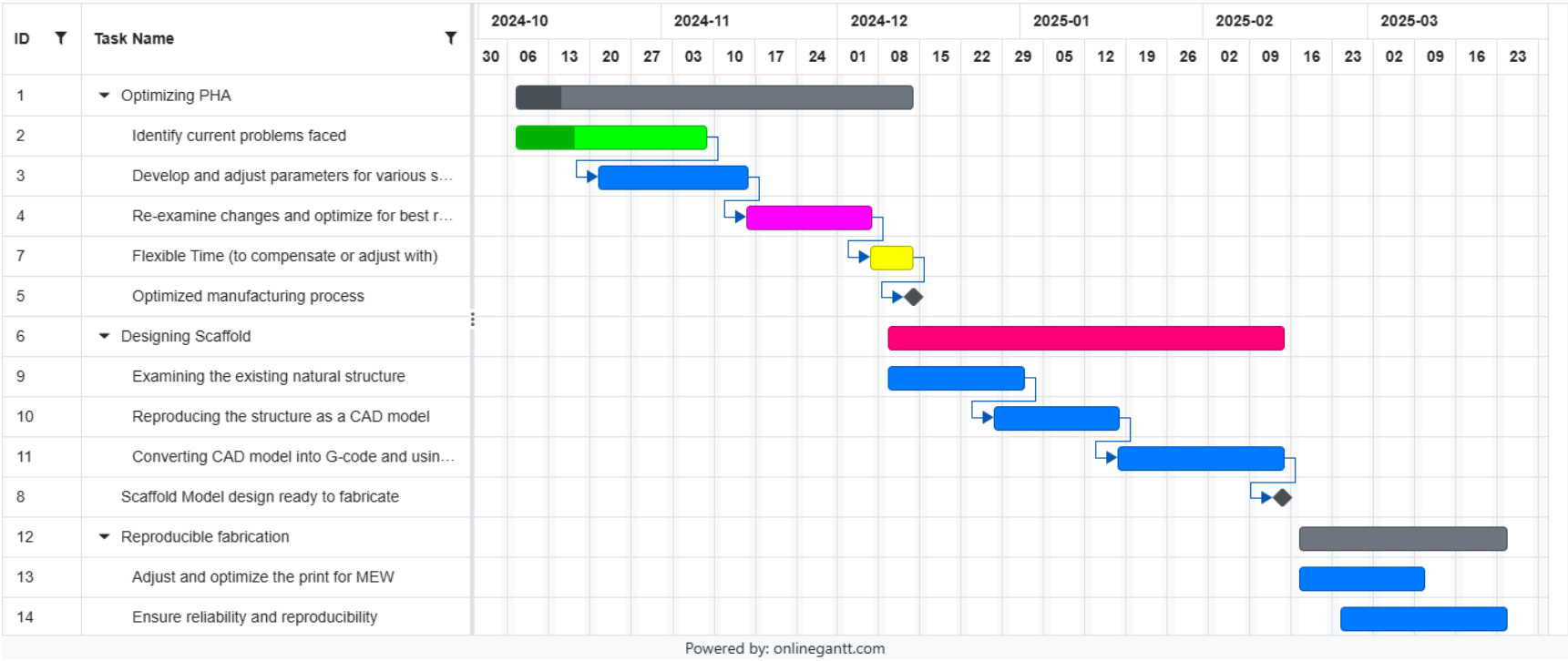
9. Appendices

9.1 Appendix A: Timeline and Project Planning

9.1.1 Initial Timeline



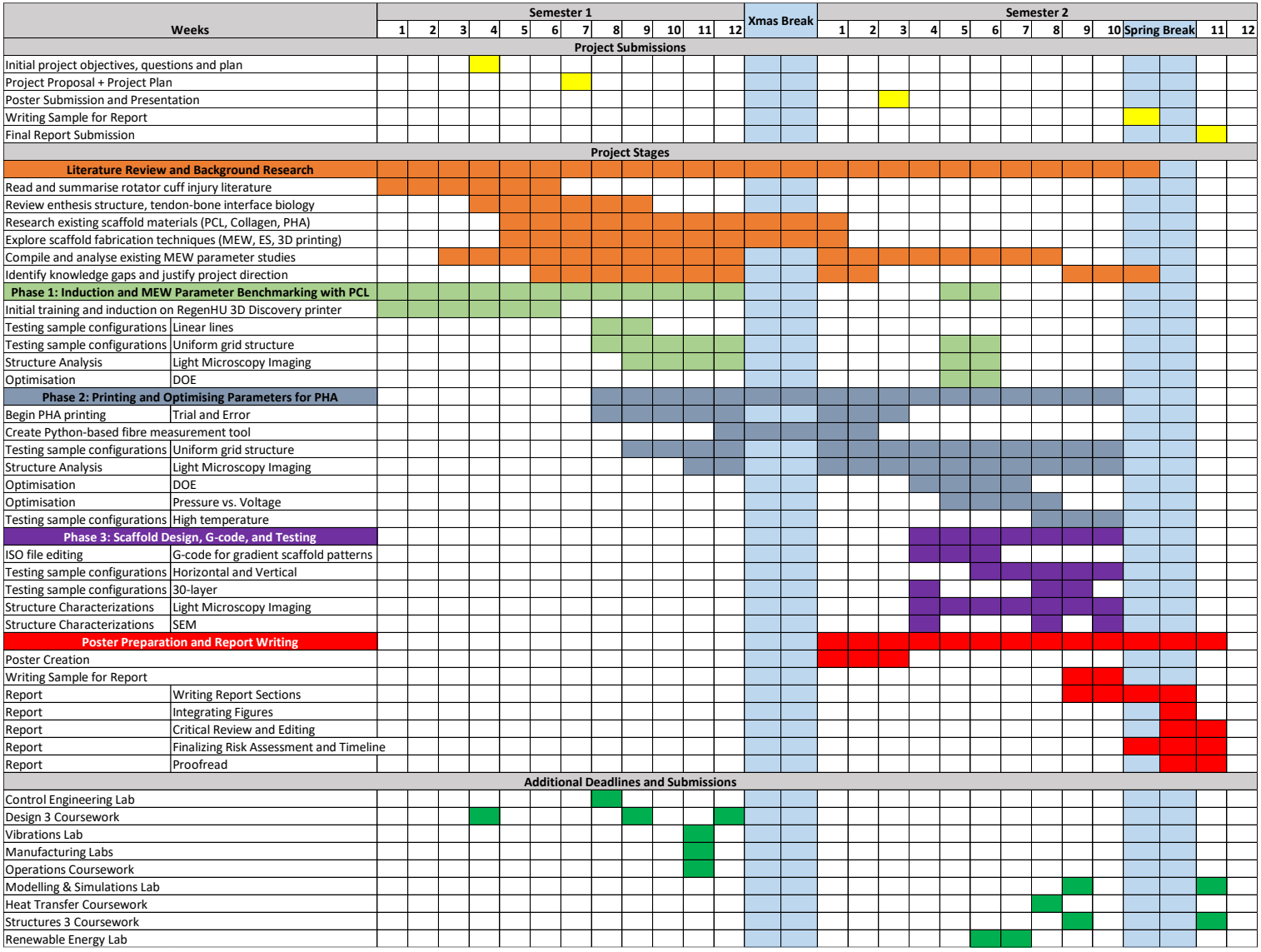
9.1.2 Proposal Timeline

Semester 1														
Weeks		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**Christmas Break**
Project Submissions														
Initial project objectives, questions and plan														
Project Proposal + Project Plan														
Project Stages														
Literature Review														
Researching MEW and bioprinting technologies														
Understanding anatomy of rotator cuff														
Analysing the native tendon structure														
Current treatment methods for tendon tears														
Analysing scaffold technology for tissue regeneration														
Investigating PHA, Collagen, and other materials														
MEW Parameter Optimization for PHA Processing														
Identify processability window														
Testing sample configurations	Lines													
Testing sample configurations	Uniform grid structure													
Structure Characterizations	Light Microscopy Imaging													
Structure Characterizations	Structural Analysis													
Additional Deadlines and Submissions														
Control Engineering Lab														
Design 3														
Vibrations Lab														
Manufacturing Labs														
Operations Coursework														

9.1.3 Proposal Feedback Timeline

Weeks		Semester 1												Xmas Break		Semester 2										Spring Break			
		1	2	3	4	5	6	7	8	9	10	11	12			1	2	3	4	5	6	7	8	9	10			11	12
Project Submissions																													
Initial project objectives, questions and plan																													
Project Proposal + Project Plan																													
Poster Submission and Presentation																													
Writing Sample for Report																													
Final Report Submission																													
Project Stages																													
Literature Review and Background Research																													
Read and summarise rotator cuff injury literature																													
Review enthesis structure, tendon-bone interface biology																													
Research existing scaffold materials (PCL, Collagen, PHA)																													
Explore scaffold fabrication techniques (MEW, ES, 3D printing)																													
Compile and analyse existing MEW parameter studies																													
Identify knowledge gaps and justify project direction																													
Phase 1: Induction and MEW Parameter Benchmarking with PCL																													
Initial training and induction on RegenHU 3D Discovery printer																													
Testing sample configurations	Linear lines																												
Testing sample configurations	Uniform grid structure																												
Structure Analysis	Light Microscopy Imaging																												
Optimisation	DOE																												
Phase 2: Printing and Optimising Parameters for PHA																													
Begin PHA printing	Trial and Error																												
Create Python-based fibre measurement tool																													
Testing sample configurations	Uniform grid structure																												
Structure Analysis	Light Microscopy Imaging																												
Optimisation	DOE																												
Optimisation	New trial runs from DOE																												
Phase 3: Scaffold Design, G-code, and Testing																													
ISO file editing	G-code for gradient scaffold patterns																												
Testing sample configurations	Horizontal and Vertical																												
Testing sample configurations	30-layer																												
Structure Characterizations	Light Microscopy Imaging																												
Structure Characterizations	SEM																												
Additional Deadlines and Submissions																													
Control Engineering Lab																													
Design 3 Coursework																													
Vibrations Lab																													
Manufacturing Labs																													
Operations Coursework																													
Modelling & Simulations Lab																													
Heat Transfer Coursework																													
Structures 3 Coursework																													
Renewable Energy Lab																													

9.1.4 Final Timeline



9.1.5 Reflection

After receiving my proposal feedback, and discussing on realistic timeframes with my supervisor, modifications were made to better encapsulate the project direction and establish milestones along a realistic progression rate. However, this approach met with complications due to lack of planning for contingencies; scheduling of the weeks was made with success in consideration for every step. As such, when faced with obstacles that setback progress due to limited availability of lab facilities, inconsistent results with expectations, or other factors, the entire timeline needed revision. This highlights the importance of placing buffer periods to accommodate the above unexpected outcomes and be more flexible.

As an undergraduate student with units and coursework alongside this project, another crucial factor to consider is the intervention of outside responsibilities and deadlines. Simply planning out the submission deadlines does not take into consideration the time needed to complete the work, and it is easy to derail off the timeline by diverting attention to these smaller projects. By finding critical points in time clashing with outside responsibilities, and appropriately dedicating the time required to fulfil all deadlines, it became easier to plan and navigate schedules for lab work and data collection.

Alongside the aforementioned, this project developed my self-learning skills to high degree, teaching me to be more proactive and independent, which translates into confidence when tackling similar large-scale projects in the future.

9.2 Appendix B: Risk Assessment

A structured risk assessment was conducted as part of this project to ensure safe, efficient, and reproducible execution of all laboratory and computational work. The following outlines the primary risks identified, the rationale behind their significance, and the mitigation measures taken.

Equipment and Access Risk

The RegenHU 3D Discovery printer, equipped with the melt electrowriting (MEW) kit, was the only available system in the lab capable of performing the required scaffold fabrication. This presented a critical single point of failure. To reduce the risk of technical downtime, a full maintenance check was performed at the start of the project to ensure the printer was in optimal working condition. Additionally, all required equipment and inductions were completed immediately, and coordination was maintained with supervisors and lab technicians for early troubleshooting.

Resource and Time Constraints

Limited access to the lab and bioprinter necessitated careful scheduling. To address this, experiments were pre-planned and conducted in batched runs, reducing setup time and maximising data collection efficiency.

Material Variability

The project relied on consistent behaviour of polyhydroxyalkanoate (PHA) batches. However, slight variations in viscosity were observed between material preparations. To manage this, process parameters were kept consistent across comparative tests, and any abnormal prints were re-evaluated to rule out material degradation.

Experimental Complexity

Optimising MEW parameters involved managing multiple interdependent variables (e.g., voltage, temperature, pressure, speed, collector distance), making the process sensitive to minor changes. As such, controlled experiments were established, where only one parameter changes at a time during initial trials, and all settings and outcomes were documented to support systematic troubleshooting. Eventually, to reduce trial-and-error

dependency, the investigation employed a Design of Experiments (DOE) approach, supported by thorough documentation protocols.

Data Accuracy and Analysis

Initial attempts to measure fibre diameters using open-source tools (e.g., ImageJ) proved unreliable. This introduced risk to the accuracy of optimisation decisions. To mitigate this, a custom Python program was developed to analyse microscope images through thousands of measurements, and return statistically filtered diameter values, with the mode being the most frequent measurement observed, and thus the average diameter value taken.

Collaboration and Contingency Planning

Unexpected delays, such as equipment downtime or experiment failure, were managed through ongoing communication with supervisory staff through email and Microsoft Teams. Frequent updates and transparent progress reporting with supervisors and collaborators enabled timely interventions when issues arose.

As this was an experimental investigation, if the project failed to achieve full optimisation of parameters – whether due to time constraints, experimental limitations, or other factors – the structured and well-documented approach taken throughout the investigation still holds significant value. Every stage of the process, from early benchmarking with PCL to later attempts at printing with PHA, was methodically recorded, helping future investigations to review, replicate, or refine each step. This transparency ensures that the work completed can serve as a foundational reference for continued optimisation and scaffold development, even if specific experimental objectives were not fully met within the project timeframe.

9.3 Appendix C: Experiment Data Tables[2]

9.3.1 Key PCL Runs:

Run	Temperature (°C)	Pressure (MPA)	Distance (mm)	Voltage (kV)	Speed (mm/s)	Rank	Diameter (mm)
1	75	0.25	3.5	7	5	2	-
2	75	0.1	3.5	7	17.5	3	-
3	75	0.25	6	7	17.5	2	0.109773
4	75	0.25	1	7	17.5	2	0.110815
5	75	0.25	3.5	12	17.5	3	0.095026
6	75	0.25	3.5	2	17.5	0	-
7	75	0.4	3.5	7	17.5	3	0.072222
8	75	0.25	3.5	7	30	4	0.057283
9	80	0.1	3.5	7	5	3	0.076882
10	80	0.25	1	7	5	3	0.060572
11	80	0.25	6	7	5	3	0.037273
12	80	0.25	3.5	2	5	0	-
13	80	0.25	3.5	12	5	1	0.059238
14	80	0.4	3.5	7	5	4	0.018634
15	80	0.1	3.5	2	17.5	0	-
16	80	0.1	6	7	17.5	5	0.013974
17	80	0.1	3.5	12	17.5	1	0.02335
18	80	0.1	1	7	17.5	6	0.014149
19	80	0.25	1	2	17.5	0	-
20	80	0.25	3.5	7	17.5	4	0.013974
21	80	0.25	3.5	7	17.5	4	0.013974
22	80	0.25	6	12	17.5	4	0.013974
23	80	0.25	3.5	7	17.5	5	0.011644
24	80	0.25	3.5	7	17.5	4	0.013974
25	80	0.25	6	2	17.5	0	-
26	80	0.25	3.5	7	17.5	4	0.011644
27	80	0.25	3.5	7	17.5	4	0.011644
28	80	0.25	1	12	17.5	3	0.013974
29	80	0.4	3.5	12	17.5	5	0.016304
30	80	0.4	3.5	2	17.5	0	-
31	80	0.4	6	7	17.5	4	0.013974
32	80	0.4	1	7	17.5	5	0.016304
33	80	0.1	3.5	7	30	6	0.013974
34	80	0.25	1	7	30	6	0.013974
35	80	0.25	3.5	2	30	0	-
36	80	0.25	6	7	30	5	0.013974
37	80	0.25	3.5	12	30	4	0.016304
38	80	0.4	3.5	7	30	6	0.013974

[2] Only the key runs with documented diameter are represented here

39	85	0.25	3.5	7	5	4	0.018634
40	85	0.1	3.5	7	17.5	3	0.013974
41	85	0.25	3.5	12	17.5	4	0.013974
42	85	0.25	6	7	17.5	3	0.013974
43	85	0.25	3.5	2	17.5	0	-
44	85	0.25	1	7	17.5	5	0.013974
45	85	0.4	3.5	7	17.5	6	0.016304
46	85	0.25	3.5	7	30	6	0.011644

9.3.2 DOE PHA Runs:

Run	Temperature (°C)	Pressure (MPA)	Distance (mm)	Voltage (kV)	Speed (mm/s)	Rank	Diameter (mm)
1	52	0.027	5	7.4	16.7	4	0.053582
2	52	0.027	5	7	16.7	4	0.053582
3	48	0.027	5	7	16.7	2	0.048923
4	44	0.027	5	7	16.7	3	0.023294
5	50	0.027	5	6.7	16.7	3	0.018634
6	50	0.05	5	6.7	16.7	2	0.039604
7	50	0.035	5	6.7	16.7	2	0.039604
8	50	0.035	4	6.4	16.7	2	0.046593
9	50	0.02	3	6	16.7	3	0.039604
10	50	0.015	4	5	15	4	0.039603
11	50	0.05	3	4	15	4	0.025624
12	50	0.25	3	4	15	4	0.041933
13	50	0.3	3	4.5	15	5	0.037274
14	50	0.01	3	5	15	3	0.011644
15	50	0.1	3	5	15	2	0.025624
16	50	0.2	3	5.5	15	4	0.027954
17	50	0.2	3	6.5	15	4	0.034944
18	50	0.1	3	7	15	4	0.032613
19	50	0.01	3	7.5	15	3	0.044263
20	50	0.3	3	7.5	15	4	0.039603
21	50	0.05	3	8	15	3	0.020964
22	50	0.25	3	8	15	5	0.039604

9.3.3 PHA High Temperature Runs

Run	Temperature (°C)	Pressure (MPA)	Distance (mm)	Voltage (kV)	Speed (mm/s)	Rank	Diameter (mm)
23H	100	0.1	3	5	15	4	0.013974
24H	100	0.1	3	4.5	15	4	0.013974
25H	100	0.1	3.5	4.5	17.5	3	0.009319
26H	100	0.1	3.5	4.5	15	4	0.009319
27H	100	0.1	3	4.5	17.5	3	0.013974
28H	100	0.1	4	5	15	4	0.009319
29H	100	0.1	2.5	4	15	4	0.013974
30H	100	0.1	2.5	3.5	15	4	0.016304
31H	100	0.1	2.5	4	20	6	0.016308
32H30L[3]	100	0.1	2.5	4	20	6	0.023301

[3] This is the high temperature 30-layer run and its observed diameter